

EvoCracker

AI RESEARCH PROJECT

Phase 2 of 5 — The Setup (Calibration Round)

PHASE 2

The Setup

Calibration round — collecting telemetry to classify the player's baseline playstyle

Deliverable Summary

- Implemented a dedicated no-enemy calibration pass on first run.
- Captured raw telemetry suitable for later classification and GA adaptation.
- Added a clean transition from calibration into evolved combat runs.

Verification Checklist

- ✓ Calibration captures keystrokes, movement, and zone dwell.
- ✓ Profile preprocessing is defined before classification.
- ✓ Exit and transition behavior is explicitly documented.

1. What is the Calibration Round?

Since EvoCracker uses real-time player data rather than a pre-existing dataset, it requires initial data to train Generation 1 enemies. A Calibration Round occurs on Floor 1, Iteration 1.

No Enemies Spawn	The player is free to explore the environment safely.
Telemetry Active	The engine records every movement, keystroke, and decision.

Speed Boost	A temporary player speed boost makes calibration feel quick rather than empty.
Goal	Establish whether the player naturally explores every room, rushes to the exit, or uses stealth/hiding mechanics.

2. Telemetry Recorded

The GameScreen continuously monitors and records the following metrics into a PlayerProfile:

- Total Moves & Time Moving — How long the player is actively traversing the map.
- Raw Keystrokes — Key down/up events stored with timestamps, then cleaned into per-key counts.
- Movement Coordinates — Tile coordinates sampled with timestamps and capped before persistence.
- Zone Dwell Time — Time accumulated for spawn, center, exit, hazard, treasure, and slow-terrain zones.
- Path Straightness — Ratio of direct displacement to actual distance traveled.
- Exploration Rate — Unique grid tiles visited versus total grid size.
- Stealth & Hiding — Time in stealth states (e.g. Ghost Cloak) or lingering in safe zones.
- Combat Stats (Later Floors) — Attack frequency, flee frequency, and items used.

3. Exiting the Calibration Round

When the player steps on physical boundary tiles or the marked exit tile, the round ends immediately and triggers finalizeLearning(). Defeating enemies is not required — the objective is reaching the exit.

4. Preprocessing and Classification

Raw telemetry is cleaned before classification: entries are capped to recent samples, key events become keystrokeCounts, zone timers produce a dominantZone, and movement data is converted into average speed, path straightness, and exploration rate.

The processed profile is passed to classifyPlaystyle() in GeneticAlgorithm.ts, which computes a score for five distinct labels:

rusher	High speed, straight paths, low stealth.
stayer	High stealth duration, high hiding frequency.
explorer	High exploration rate (unique tiles), non-straight paths.

fighter	High engagement rate, low flee frequency (post-calibration).
hybrid	If scores are closely balanced with no dominant pattern.

5. The Transition & Result

Upon exit, a 1.5-second full-screen overlay appears: "Calibration Round Completed. Loading Level 1..."

- 1

Locks in the classified player profile.
- 2

Applies a Playstyle-Adaptive Bias to Generation 1 (e.g., rusher → enemies spawn with higher speed and an A* algorithm bias).
- 3

Smoothly reloads the exact same Tiled Battleground map, now internally on Floor 2.
- 4

When the loading screen disappears, the player begins Iteration 2 (Level 1) and Generation 1 enemies instantly spawn.